

ROBOT EQUILIBRIUM CONDITIONS

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A robot R is in equilibrium if and only if all its links $\{l_i\}$ are in equilibrium then

$$\begin{cases} \underline{F}_i = \underline{0} \\ \underline{M}_i^c = \underline{0} \end{cases}$$

$\uparrow$
N-E eq.s

$$i = 1 \div n$$

$\uparrow$
nr. links



$$\begin{cases} \underline{F}_i = \underline{0} \\ \underline{M}_i = \underline{0} \end{cases}$$

$$i = 1 \div n$$

(6x n eq.s)

$\uparrow$
moments computed at P_i ($M_i = M_i^c$ since $\underline{F}_i = \underline{0}$)

Compute the virtual work of each f/t (mult. ply by $\delta \underline{P}_i, \delta \theta_i$)

$$\begin{cases} \underline{F}_i \cdot \delta \underline{P}_i = 0 \\ \underline{M}_i \cdot \delta \theta_i = 0 \end{cases}$$

$$i = 1 \div n$$

(2x n eq.s.)

Add up all the eqs. above

$$\sum_{i=1}^n \underline{F}_i \cdot \delta \underline{P}_i + \underline{M}_i \cdot \delta \theta_i = 0 \quad \sim \text{total virtual work}$$

Define $\underline{F}_i = \underset{\substack{\uparrow \\ \text{active}}}{\underline{F}_i^{(A)}} + \underset{\substack{\uparrow \\ \text{reactive}}}{\underline{F}_i^{(R)}}$

$$\underline{M}_i = \underline{M}_i^{(A)} + \underline{M}_i^{(R)}$$

Then

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$$\begin{aligned} \delta A &= \sum_i^n [\underline{F}_i^{(A)} + \underline{F}_i^{(R)}] \cdot \underline{\delta P}_i + [\underline{M}_i^{(A)} + \underline{M}_i^{(R)}] \cdot \underline{\delta \theta}_i = \\ &= \sum_i^n (\underline{F}_i^{(A)} \underline{\delta P}_i + \underline{M}_i^{(A)} \underline{\delta \theta}_i) + \underbrace{\sum_i^n (\underline{F}_i^{(R)} \underline{\delta P}_i + \underline{M}_i^{(R)} \underline{\delta \theta}_i)}_{=0} = 0 \end{aligned}$$

virtual work principle

Recall that :

$$\begin{cases} \underline{F}_i^{(A)} = \underline{f}_i^{ACT} + \underline{F}_i^{EXT} \\ \underline{M}_i^{(A)} = \underline{m}_i^{ACT} + \underline{M}_i^{EXT} \end{cases}$$

Then

$$\underbrace{\sum_i^n \underline{f}_i^{ACT} \underline{\delta P}_i + \underline{m}_i^{ACT} \underline{\delta \theta}_i}_{\text{total virtual work of (conjugate) actuation f/t}} + \underbrace{\sum_i^n \underline{F}_i^{EXT} \underline{\delta P}_i + \underline{M}_i^{EXT} \underline{\delta \theta}_i}_{\text{total virtual work of external f/t}}$$

$$\delta \underline{q}^T \left[\sum_i^n \overset{\circ}{J}_{i/0}^{LT} \underline{F}_i^{EXT} + \sum_i^n \overset{\circ}{J}_{i/0}^{NT} \underline{M}_i^{EXT} \right]$$

$\delta \underline{q}^T \underline{I}$

Then the following equality must hold

$$\delta \underline{q}^T \left[\underline{I} + \sum_i^n \overset{\circ}{J}_{i/0}^{LT} \underline{F}_i^{EXT} + \sum_i^n \overset{\circ}{J}_{i/0}^{NT} \underline{M}_i^{EXT} \right] = 0$$

$\delta \underline{q}$

Which can also be rewritten as

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$$\underline{\delta q}^T \left[\underline{I} + \sum_i^n {}^0 \underline{J}_{i/o}^T \begin{pmatrix} {}^0 \underline{M}_i^{\text{EXT}} \\ {}^0 \underline{F}_i^{\text{EXT}} \end{pmatrix} \right] = 0 \quad \forall \underline{\delta q}$$

NOTE Let $\underline{\delta q}$ be an arbitrary (infinitesimal) vector

$$\text{if } \underline{\delta q}^T \underline{x} = 0 \quad \forall \underline{\delta q} \Rightarrow \underline{x} = 0$$

In fact consider $\underline{\delta q}^T = [0 \dots \delta q_i \dots 0]$, $\delta q_i \neq 0$, $\underline{\delta q}^T \underline{x} = \delta q_i x_i = 0 \Rightarrow x_i = 0 \quad \forall i$

Then at the equilibrium the following condition must hold

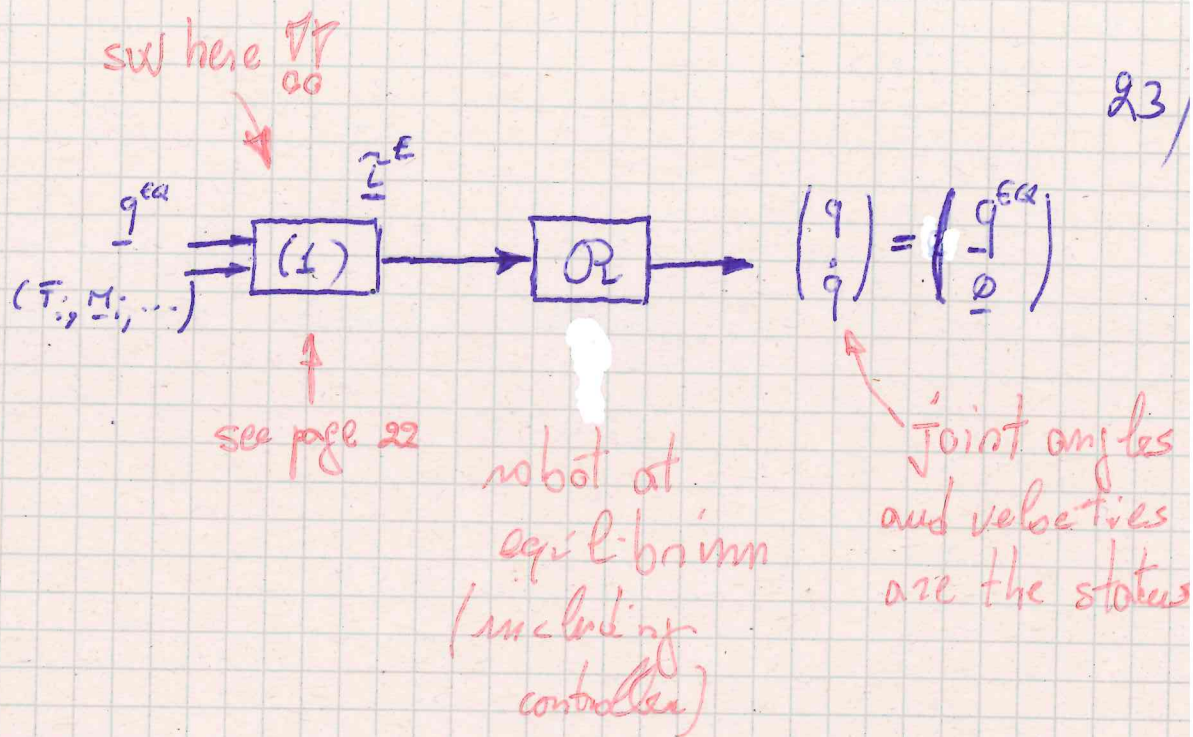
$$\underline{I} + \sum_i^n {}^0 \underline{J}_{i/o}^T \begin{pmatrix} {}^0 \underline{M}_i^{\text{EXT}} \\ {}^0 \underline{F}_i^{\text{EXT}} \end{pmatrix} = 0$$

NOTE Given a fully actuated robot (*) then

$$\underline{I}^{\text{EQ}} = - \sum_i^n {}^0 \underline{J}_{i/o}^T \begin{pmatrix} {}^0 \underline{M}_i^{\text{EXT}} \\ {}^0 \underline{F}_i^{\text{EXT}} \end{pmatrix} \quad (1)$$

is the (nominal) set of generalized joint f/t required to guarantee the equilibrium condition of the robot

*) featuring joint congruent actuator (what happens if the actuators are not joint congruent?)



Important

Note (*) Recall that ${}^0\bar{J}_{c_i/o} = {}^0S_{c_i/2} {}^0\bar{J}_{i/o}$ hence

$${}^0\bar{J}_{i/o} = {}^0S_{c_i/2}^{-1} {}^0\bar{J}_{c_i/o}$$

↑
 ${}^0S_{i/c_i}$

Then:

$${}^0\bar{J}_{i/o}^T \begin{bmatrix} {}^0\bar{M}_i^{ext} \\ {}^0\bar{F}_i^{ext} \end{bmatrix} = {}^0\bar{J}_{c_i/o}^T \underbrace{{}^0S_{i/c_i}^T \begin{bmatrix} {}^0\bar{M}_i^{ext} \\ {}^0\bar{F}_i^{ext} \end{bmatrix}}_{\substack{\text{f/t expressed} \\ \text{at } P_i}} = {}^0\bar{J}_{c_i/o}^T \begin{bmatrix} {}^0\bar{M}_{c_i}^{ext} \\ {}^0\bar{F}_{c_i}^{ext} \end{bmatrix}$$

*) This note is important and valid besides the eq. conts.

Following the last note the eq. condition can be rewritten as

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$$\underline{\underline{I}} + \sum_i^m {}^0\bar{J}_{C_i/o}^T \begin{bmatrix} {}^0\bar{M}_{C_i}^{ext} \\ \bar{F}_{C_i}^{ext} \end{bmatrix} = \underline{\underline{0}} \quad (2)$$

Now recall that gravity cannot be switched off then it is usual to separate the contribution of the gravitational forces from the possible other external f/t (usually due to contacts or other interactions of the robot with the environment)

$$\begin{aligned} \bar{M}_{C_i}^{ext} &\Leftrightarrow \bar{M}_{C_i}^{ext} \\ \bar{F}_{C_i}^{ext} &\Leftrightarrow \bar{F}_{C_i}^{ext} + m_i \underline{g} \end{aligned}$$

$\uparrow$
 mass

or "change of n variables"

Eq. (2) can be rewritten as:

$$\underline{\underline{I}} + \sum_i^m {}^0\bar{J}_{C_i/o}^T \begin{bmatrix} {}^0\bar{M}_{C_i}^{ext} \\ \bar{F}_{C_i}^{ext} \end{bmatrix} + \underbrace{\sum_i^m {}^0\bar{J}_{C_i/o}^T m_i \underline{g}}_{-G(q)} = \underline{\underline{0}}$$

$-G(q)$

the Jacobians are functions of the robot configuration and then of the joint angles.

RELEVANT CASES

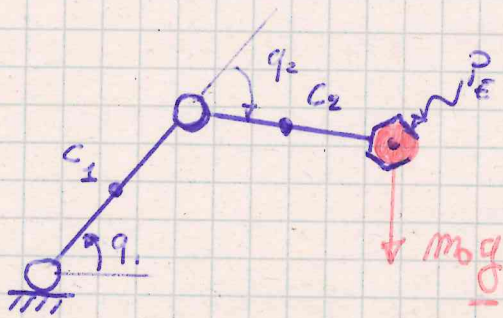
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① No external f/t

$$\underline{\tau}^{eq} = C(q) = - \sum_{i=1}^n {}^0\dot{\underline{J}}_{e_i}^T m_i \underline{g}$$

② Robot holding an object of mass m_o at the end-effector

$$\underline{\tau}^{eq} = G(q) - {}^0\dot{\underline{J}}_{c_n}^T S_{p/c_n}^T (m_o \underline{g})$$



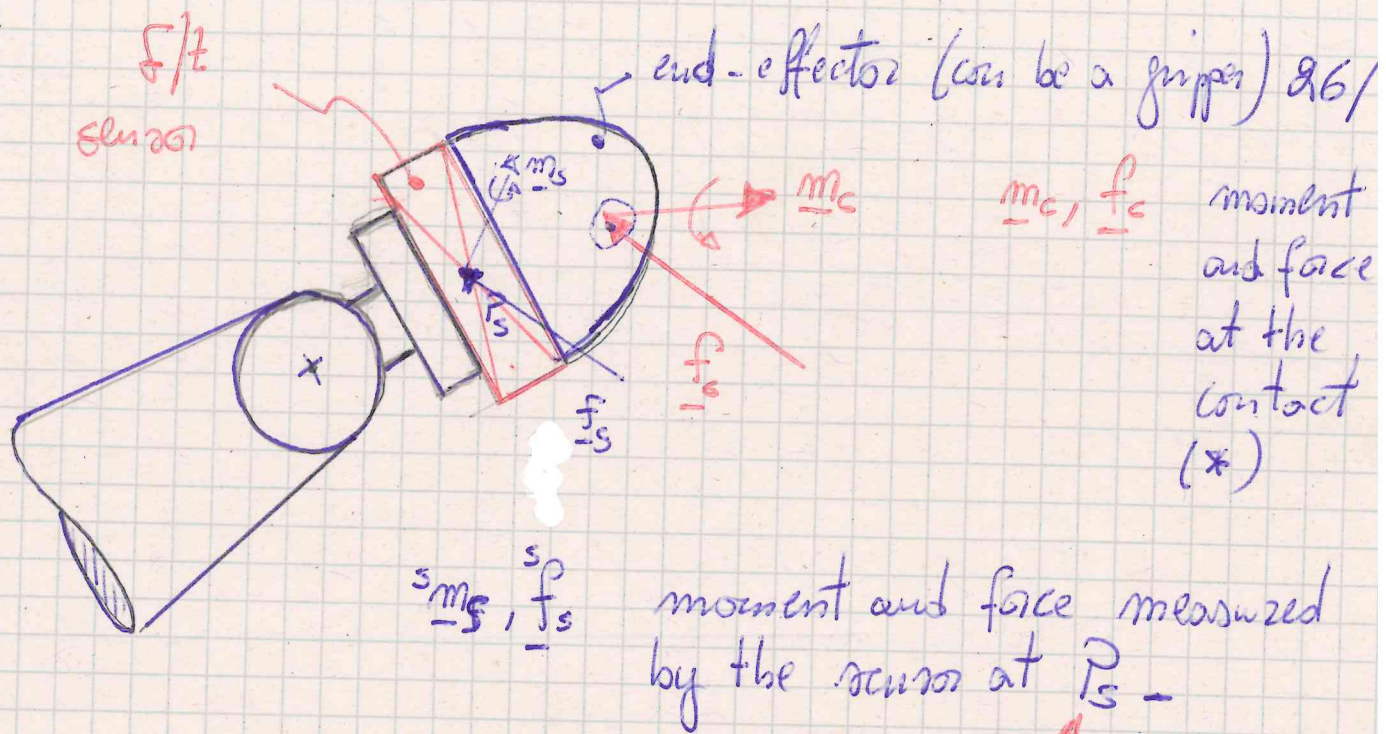
③ Robot "reacting" to a f/t at the end-effector

$$\underline{\tau}^{eq} = G(q) - {}^0\dot{\underline{J}}_{c_n}^T S_{p/c_n}^T \begin{pmatrix} {}^0\underline{M}_e \\ {}^0\underline{T}_e \end{pmatrix}$$

NOTE In the general case of interaction of the robot with the environment (objects, humans etc.) it is necessary to know:

$\underline{M}_e, \underline{T}_e$
 ↑ ↑
 resultant of
 forces and moments

← if the contact is certainly at the end effector the robot can be equipped with a 6-axis f/t sensor.



the position of P_s w.r.t. P_c (or P_n , or C_n) is part of the sensor calibration (**)

NOTE Some robots are equipped with joint torque (force) sensors. These sensors allow to measure the projection in joint space of forces/torques, but also inertial effects (see later). In the static case $\underline{T}_m$ the vector of measured joint torques could be used to compute $\underline{\tau}_{ext}$ (although the things might not be so straightforward)

**) calibration, parameter identification are not covered here

*) They might be distributed